

Market Power in Indian Organised Manufacturing: Trends, Determinants, and Labour Market Implications

Laxmikanta Sarangi* Falguni Pattanaik†

May 2026

Abstract

This paper estimates product market markups for Indian organised manufacturing using plant-level data from the Annual Survey of Industries (ASI) for 2012–13 to 2023–24 ($N = 515,374$ plant-year observations; 79,704 unique factories across 23 two-digit NIC industries). Markups are estimated via the production-function approach of De Loecker and Warzynski (2012), combining Cobb-Douglas specification with the Akerberg–Caves–Frazer (ACF) proxy-variable estimator (Akerberg et al., 2015). Contrary to the global trend of rising markups documented by De Loecker et al. (2020), we find that the sales-weighted mean markup in Indian organised manufacturing *fell* from 1.88 in 2012–13 to 1.73 in 2023–24 (a decline of 8.2%), though it remains substantially above unity throughout. The median markup is more stable (1.20 in 2013 to 1.17 in 2024), indicating that the mean is shaped by a fat right tail of high-markup plants. Markup heterogeneity across industries is large: median markup ranges from 2.00 in tobacco products (NIC2 12) to 1.07 in basic metals (NIC2 24). Within-plant regressions show that larger output market share raises market power ($\hat{\delta} = 11.16$, $p < 0.001$), while larger employment share reduces

*Corresponding author of PhD Student, Dept. of HSS (Economics), IIT Roorkee, Roorkee 247667, Uttarakhand, India. *Email:* `laxmikanta_s@hs.iitr.ac.in`.

†Associate Professor, Dept. of HSS (Economics), IIT Roorkee, Roorkee 247667, Uttarakhand, India.

the labour markdown ($\hat{\gamma} = -3.24$, $p < 0.001$), consistent with dynamic monopsony. Higher markup is associated with lower wages ($\hat{\beta} = -0.0208$, $p < 0.001$), and higher markdown is similarly associated with lower wages ($\hat{\beta}(\psi) = -0.0014$, $p < 0.001$). The labour share of revenue has fallen from 22.3% in 2012–13 to 16.2% in 2023–24. These findings have direct implications for competition policy and labour market regulation in India.

JEL codes: D22, D43, J42, L11, O14.

Keywords: markup, markdown, monopsony, market power, Indian manufacturing, ASI, production function, Cobb-Douglas.

1 Introduction

The study of firm-level market power has generated substantial attention in macroeconomics and industrial organisation following the seminal work of Hall (1988) and the more recent contributions of De Loecker and Warzynski (2012) and De Loecker et al. (2020). De Loecker et al. (2020) document a striking rise in average markups for publicly listed US firms from 1.21 in 1980 to 1.61 in 2016, and find broadly similar trends in other advanced economies. Rising markups compress workers’ share of revenue, raise product prices, and distort resource allocation (Manning, 2003).

India presents a particularly important case. Despite policy initiatives such as Make in India (2014) and implementation of the Goods and Services Tax (GST, 2017), organised manufacturing’s share of GDP has stagnated near 15–17%. The labour share of income in organised manufacturing has declined persistently (De Loecker et al., 2020), raising concerns about whether the sector’s growth has been accompanied by rising market power that benefits firm owners at the expense of workers. Yet direct, plant-level evidence on markup trends in Indian manufacturing over the post-2012 period remains scarce.

This paper fills that gap. Using Annual Survey of Industries (ASI) unit-level data for 2012–13 to 2023–24 ($N = 515,374$ plant-year observations across 79,704 unique establishments), we estimate firm-level product market markups via the production-

function approach of De Loecker and Warzynski (2012), combined with the ACF proxy-variable estimator (Akerberg et al., 2015). We additionally recover a measure of labour market power (markdown) following Brooks et al. (2021), and examine how markups and markdowns relate to wages at the plant level.

Our main findings are as follows. First, the sales-weighted mean markup fell from 1.88 in 2012–13 to 1.73 in 2023–24, a decline of 8.2%. This *contrasts* with the global rising-markup narrative but is consistent with increased import competition and formalisation pressures following GST. Second, markup heterogeneity across industries and states is substantial: tobacco products and non-metallic mineral products (NIC2 12 and 23) command median markups of 2.00 and 1.63 respectively, while food products and basic metals (NIC2 10 and 24) are near unity (1.07–1.08). Third, within-plant identification shows that output-market share and employment-market share are significant determinants of markup and markdown respectively, consistent with both product and labour market power. Fourth, both higher markup and higher markdown are associated with lower wages, confirming that market power extracts rents at the expense of workers.

The paper contributes to the literature in three ways. First, it provides the most comprehensive plant-level markup estimates for Indian organised manufacturing using 12 years of ASI data. Second, it jointly estimates output markup and labour markdown, linking both to wages within a unified framework. Third, it documents a *falling* markup trend in a major emerging economy, offering a counterpoint to the advanced-economy evidence of De Loecker et al. (2020).

The remainder of the paper is organised as follows. Section 2 surveys the related literature. Section 3 describes the data. Section 4 presents the methodology. Section 5 reports the main results. Section 6 examines robustness. Section 7 discusses economic implications. Section 8 concludes.

2 Related Literature

2.1 Measuring Markups from Production Functions

The production-function approach to estimating firm-level markups traces to Hall (1988), who showed that markups can be inferred from the ratio of the output elasticity of a variable input to its observed revenue share. De Loecker and Warzynski (2012) operationalised this as a plant-level estimator using the cost-minimisation first-order condition, allowing markups to vary across plants and over time without requiring price data. Consistent estimation of output elasticities is complicated by the endogeneity of input choices: firms adjust inputs in response to productivity shocks that the econometrician does not observe (Levinsohn and Petrin, 2003). Akerberg et al. (2015) resolve the collinearity problem present in earlier proxy-variable estimators and provide the theoretical foundation for the LP+ACF approach used in this paper. Gandhi et al. (2020) establish non-parametric identification of output elasticities in gross-output specifications under standard assumptions, validating the approach for settings like ours where intermediate inputs are an important component of production. Syverson (2025) provides a recent comprehensive survey of the markup and markdown literature, covering measurement methodologies, distributional trends across countries, and the macroeconomic implications of rising market power, offering the broader context within which our Indian evidence sits.

2.2 Global Evidence on Market Power Trends

The empirical literature on market power has documented a broad rise in markups in advanced economies over recent decades. De Loecker et al. (2020) find that the sales-weighted mean markup of publicly listed US firms rose from 1.21 in 1980 to 1.61 in 2016, with similar though smaller increases in Europe. Autor et al. (2020) link this trend to the rise of “superstar firms” — high-productivity, high-market-share enterprises — and connect it to the concurrent decline of the labour share of income in the United States. Barkai (2020) documents simultaneous declines in both the labour and capital shares of US income since the 1980s, attributing the residual to rising pure profits consistent with

increasing product market power. Karabarbounis and Neiman (2014) document a global decline in the labour share across more than 50 countries and most sectors since the early 1980s, linking it partly to falling relative prices of investment goods; rising product market power has since been offered as a complementary explanation (Syverson, 2019). These findings motivate our investigation of whether the same forces are operating in Indian organised manufacturing, though we document a *contrasting* trend over our sample period.

2.3 Labour Monopsony and the Markdown

On the labour side, Manning (2003) provides the foundational theory of dynamic monopsony: when workers face search frictions and firms have wage-setting power, each firm faces an upward-sloping labour supply curve and pays wages below the value of the marginal product of labour. Manning (2021) updates this framework with a comprehensive review of the empirical monopsony literature, concluding that estimates of the labour supply elasticity to the firm typically lie between 1 and 4 — far below the competitive benchmark of infinity — indicating widespread wage-setting power across industries and countries. A meta-analysis by Sokolova and Sørensen (2021) of over 1,300 estimates drawn from 67 studies corroborates this finding: the average labour supply elasticity to the firm is approximately 1.5, implying that wages are roughly 40% below the marginal revenue product. Berger et al. (2022) develop and estimate a general equilibrium model of labour market power using US data, finding that labour market concentration generates significant welfare losses and that wages fall substantially below competitive levels even in markets with many employers. Brooks et al. (2021) formalise a production-data measure of labour market power — the *markdown* $\psi = \mu^L / \mu^M$ — that can be estimated from the same data used for product markup estimation. They apply this framework to Indian ASI data for 1998–2013, finding widespread monopsony (median ψ substantially above unity) and a negative relationship between employment market share and the markdown, consistent with dynamic monopsony. Hsieh and Klenow (2009) document large misallocation of resources across firms in India and China; market power in both product and labour markets is a plausible driver of the distortions in their decomposition.

2.4 Indian Manufacturing: Productivity, Trade, and Market Structure

Research on Indian manufacturing has accelerated since the trade liberalisation reforms of 1991. Goldar (2004) and Das (2004) document productivity improvements in the 1990s following trade reform and delicensing. Goldberg et al. (2010) show that access to imported intermediate inputs raised domestic product variety and firm-level productivity, suggesting that input market liberalisation can compress markups by reducing cost asymmetries. For our period (2013–2024), two structural changes are particularly relevant: demonetisation (November 2016) and the implementation of the Goods and Services Tax (July 2017). GST unified India’s fragmented state-level indirect tax structure, eliminated cascading taxes on interstate transactions, and effectively deepened cross-state competition, especially for goods produced by large plants in nationally traded industries. Brooks et al. (2021) provide the only prior comprehensive study of both markups and markdowns for Indian organised manufacturing, but their sample ends in 2013. The present paper extends the analysis by a decade, spanning the GST transition and the COVID-19 shock, and examines whether the monopsony patterns they document have persisted or changed.

3 Data

3.1 Annual Survey of Industries

The primary data source is the Annual Survey of Industries (ASI), conducted annually by the Ministry of Statistics and Programme Implementation (MOSPI), Government of India. The ASI covers all registered manufacturing establishments under the Factories Act, 1948 — specifically, factories employing 10 or more workers using power, or 20 or more workers without power. Each factory is identified by a unique Dispatch Serial Number (DSL). The survey comprises ten data blocks (A through J) recording factory identification, employment, fixed assets, working capital, expenses, and product details.

We use data for fiscal years 2012–13 to 2023–24 (year codes 2013–2024 throughout),

covering 12 annual waves. After merging blocks, applying cleaning rules, and restricting to manufacturing NIC2 industries 10–33, the estimation sample consists of 515,374 plant-year observations (≈ 5.15 lakh) across 79,704 unique factories, 23 two-digit NIC industries, and up to 37 states and union territories.

Panel structure. This is an *unbalanced panel*: each of the 79,704 unique factories (identified by a unique Dispatch Serial Number, DSL) may appear in between one and all twelve survey waves, giving 515,374 factory-year observations in total — not 5.15 lakh unique establishments. On average, each factory appears in approximately 6.5 of the 12 waves. The unbalanced structure arises from the ASI’s two-stratum sampling design. *Census-sector* plants (factories employing 100 or more workers, or those flagged as Census units by MOSPI) are mandatory inclusions and are surveyed every year; their sampling multiplier equals one. *Sample-sector* plants (smaller factories below the Census threshold) are drawn by stratified random sampling within each state $\times$ NIC4 cell; they appear in the survey on a rotating basis with multipliers greater than one that reflect the inverse of their selection probability. All descriptive statistics and regressions use the ASI sampling multiplier as an analytical weight to restore population representativeness.

3.2 Variable Construction

Output (Y). Gross value of output from Block J (ex-factory value of products and by-products), deflated to 2011–12 prices using industry-specific Wholesale Price Index (WPI) deflators.

Materials (M). Total value of intermediate inputs consumed from Block H (indigenous inputs consumed) plus Block I (imported inputs consumed), deflated using material-specific WPI. We use *consumed* values from Blocks H and I, not Block D inventory stocks, consistent with the gross-output production function framework.

Labour (L). Total persons engaged at SNO=9 of Block E (all categories: workers, supervisory staff, and other employees). Total labour cost (W) equals wages and salaries from Block E (SNO=9) plus bonus, provident fund contributions, and welfare expenses

from Block A (emoluments in the broadest sense), following De Loecker and Warzynski (2012).

Capital (K). Closing gross fixed assets excluding land (SNO=1) and capital work-in-progress (SNO=9) from Block C, following Goldar (2004) and Das (2004). Net fixed assets (Block C, SNO=10) serve as a robustness check. Capital is deflated using an implicit capital deflator.

Expenditure shares. The material cost share $\alpha_{M,it} = (P_M M_{it}) / (P_Y Y_{it})$ and labour cost share $\alpha_{L,it} = (W_{it}) / (P_Y Y_{it})$ are computed at plant-year level from nominal values.

3.3 Summary Statistics

Table 1 presents weighted summary statistics (weights = ASI sampling multipliers) for the full estimation sample.

Table 1: Summary Statistics (Weighted by ASI Sampling Multiplier)

Variable	Mean	Median	Std. dev.	N
<i>Panel A: Market power measures</i>				
Output markup μ_{it}^M	1.81	1.19	3.46	515,374
Labour markdown ψ_{it}	4.37	2.01	6.75	515,066
Inverse markup $1/\mu_{it}^M$	0.84	—	0.40	515,374
<i>Panel B: Plant characteristics</i>				
Total persons engaged	94.1	25.0	278.2	515,374
Material cost share α_M	0.693	0.720	—	515,374
Labour cost share α_L	0.193	—	—	515,374
Output (Rs. million, nominal)	473	51	5,570	515,374
Wage bill (Rs. million, nominal)	24.4	3.6	100.5	515,374

Note: Plant-year observations, NIC2 10–33, FY2012–13 to FY2023–24. Statistics are weighted by ASI sampling multiplier. Markup and markdown winsorised 2/98 within NIC2×year.

4 Methodology

4.1 Production-Function Markup Estimator

Following De Loecker and Warzynski (2012), the firm-level output markup is derived from the first-order condition of cost minimisation with respect to a variable input. Let the production function be

$$Y_{it} = F(V_{it}, K_{it}; \omega_{it}) \cdot \varepsilon_{it}, \quad (1)$$

where V_{it} is a variable input (materials or labour), K_{it} is capital (quasi-fixed), ω_{it} is Hicks-neutral productivity, and ε_{it} is an i.i.d. measurement error. Cost minimisation yields

$$\mu_{it} = \frac{\theta_{it}^V}{\alpha_{it}^V}, \quad (2)$$

where $\theta_{it}^V = \partial \ln Y_{it} / \partial \ln V_{it}$ is the output elasticity of the variable input and $\alpha_{it}^V = P_{it}^V V_{it} / P_{it}^Y Y_{it}$ is its revenue expenditure share. With materials as the variable input, equation (2) identifies the output markup μ_{it}^M . With labour, it identifies the labour markup $\mu_{it}^L = \theta_{it}^L / \alpha_{L,it}$.

4.2 Labour Markdown

Following Brooks et al. (2021), the labour markdown (measure of wage-setting power) is

$$\psi_{it} = \frac{\mu_{it}^L}{\mu_{it}^M} = \frac{\theta_{it}^L / \alpha_{L,it}}{\theta_{it}^M / \alpha_{M,it}}. \quad (3)$$

When $\psi_{it} > 1$, the firm pays a wage below the value of the marginal product of labour — evidence of monopsony power. $\psi_{it} = 1$ corresponds to a competitive labour market.

4.3 Cobb-Douglas Production Function and ACF Estimation

The production function is specified in Cobb-Douglas log-linear form:

$$\ln Y_{it} = \beta_L \ln L_{it} + \beta_M \ln M_{it} + \beta_K \ln K_{it} + \omega_{it} + \varepsilon_{it}, \quad (4)$$

where all inputs are in real terms. In the Cobb-Douglas specification, the output elasticities β_L , β_M , β_K are industry-specific constants (estimated separately for each NIC2 industry). The plant-level markup is then $\mu_{it}^M = \beta_M^i / \alpha_{M,it}$, where β_M^i is the industry elasticity and $\alpha_{M,it}$ varies at plant-year level.

We employ the Akerberg–Caves–Frazer (ACF) proxy-variable estimator (Akerberg et al., 2015), which corrects the collinearity problem in earlier LP estimators (Levinsohn and Petrin, 2003). Materials M_{it} serve as a proxy for unobserved productivity ω_{it} in a first stage. The second stage identifies β_L and β_K via GMM moment conditions using lagged capital and lagged labour as instruments for the productivity-input covariance. Estimation is conducted separately for each two-digit NIC industry. Bootstrap standard errors use 50 replications.

Note on Translog specification. A Translog production function (with nine elasticity parameters) was tested for all 23 NIC2 industries. However, more than 98% of ASI factories are observed in only one or two years in the panel, providing insufficient within-plant variation to identify the nine quadratic and cross-product coefficients. The Translog estimates yield implausible returns-to-scale near 1.9 and unstable industry-level elasticities. This finding is consistent with Brooks et al. (2021), who similarly adopt the Cobb-Douglas specification on ASI-type data and explicitly note the identification problems with ACF Translog in short panels. Accordingly, the Cobb-Douglas specification is our primary and only reported specification.

4.4 Markup Aggregation and Trend Analysis

The aggregate markup for year t is the sales-weighted mean:

$$\bar{\mu}_t = \sum_i \frac{P_{it}^Y Y_{it}}{\sum_j P_{jt}^Y Y_{jt}} \cdot \mu_{it}. \quad (5)$$

In practice, we use the ASI sampling multiplier as the weight (which captures the sampling design) rather than the nominal output share.

4.5 Regression Specifications

Output market power. Following De Loecker and Warzynski (2012), we test whether firms with larger market share command higher markups:

$$(1/\mu_{it}^M) = \alpha_i + \lambda_t + \delta \cdot s_{it}^{out} + \varepsilon_{it}, \quad (6)$$

where s_{it}^{out} is the plant’s output share in its NIC2×year market, α_i is a plant fixed effect, and λ_t is a year fixed effect. A positive $\hat{\delta}$ means larger market share is associated with lower inverse markup (higher markup).

Labour market power.

$$\psi_{it} = \alpha_i + \lambda_t + \gamma \cdot s_{it}^{emp} + \varepsilon_{it}, \quad (7)$$

where s_{it}^{emp} is the plant’s employment share in its state×NIC2×year labour market. A negative $\hat{\gamma}$ is consistent with dynamic monopsony: as a firm expands its labour market share, it must raise wages, compressing its markdown.

All regressions use plant fixed effects absorbed via `areg`, ASI sampling multiplier as analytical weights, and standard errors clustered at the plant level.

5 Results

5.1 Production Function Estimates

Table 2 presents the industry-level Cobb-Douglas output elasticities estimated by LP+ACF, together with implied returns to scale ($RTS = \beta_L + \beta_M + \beta_K$) and mean markup.

Returns to scale are close to unity for most industries (range 0.92–1.10), consistent with the standard maintained assumption of approximately constant returns in the industry-level ACF literature (De Loecker and Warzynski, 2012). Materials account for the largest share of the output elasticity in most industries, as expected for the gross-output specification. Pharmaceuticals (NIC2 21) and non-metallic minerals (NIC2 23) are notable exceptions,

Table 2: Cobb-Douglas Production Function Estimates by NIC2 Industry

NIC2	Industry	$\hat{\beta}_M$	$\hat{\beta}_L$	$\hat{\beta}_K$	RTS	Mean μ^M
10	Food products	0.878	0.153	0.013	1.044	1.246
11	Beverages	0.471	0.561	0.073	1.104	7.938
12	Tobacco products	0.837	0.189	0.034	1.060	2.356
13	Textiles	0.888	0.141	-0.009	1.020	1.498
14	Wearing apparel	0.795	0.167	0.037	0.999	1.690
15	Leather & products	0.825	0.158	0.017	1.000	1.357
16	Wood & wood products	0.800	0.181	0.049	1.030	1.251
17	Paper & paper products	0.759	0.113	0.070	0.942	1.267
18	Printing & reproduction	0.759	0.113	0.070	0.942	1.370
19	Coke & petroleum	0.867	0.134	0.005	1.005	1.208
20	Chemicals	0.730	0.255	0.067	1.053	1.430
21	Pharmaceuticals	0.649	0.426	0.012	1.087	2.039
22	Rubber & plastics	0.769	0.185	0.031	0.985	1.283
23	Non-metallic minerals	0.555	0.458	0.075	1.087	3.880
24	Basic metals	0.815	0.169	0.037	1.021	1.217
25	Fabricated metals	0.769	0.208	0.043	1.020	1.408
26	Electronics & optical	0.745	0.212	0.028	0.985	1.437
27	Electrical equipment	0.813	0.164	0.035	1.012	1.271
28	Machinery & equipment	0.769	0.219	0.043	1.031	1.433
29	Motor vehicles	0.667	0.235	0.019	0.921	1.361
30	Other transport	0.770	0.188	0.053	1.011	1.357
31	Furniture	0.797	0.238	0.030	1.065	1.435
32	Other manufacturing	0.711	0.189	0.074	0.974	1.374

Note: Cobb-Douglas LP+ACF estimates (prodest, 50 bootstrap reps). RTS = $\hat{\beta}_L + \hat{\beta}_M + \hat{\beta}_K$. Mean markup = weighted mean $\hat{\beta}_M/\alpha_{M,it}$. NIC 2008 two-digit classification.

Beverages (NIC2 11): mean markup 7.94 reflects a fat right tail; median markup is 1.24. Industries 17 and 18 share a prodest specification group; coefficients are as reported.

with relatively high labour elasticities.

5.2 Markup Distribution

The weighted mean markup across all plant-years is 1.81 and the weighted median is 1.19 (Table 1). The distribution is highly right-skewed (standard deviation 3.46): 68.7% of plant-years show markdown $\psi > 1$ and 47.7% show $\psi > 2$. Figure 1 shows the kernel density of log markup and log markdown.

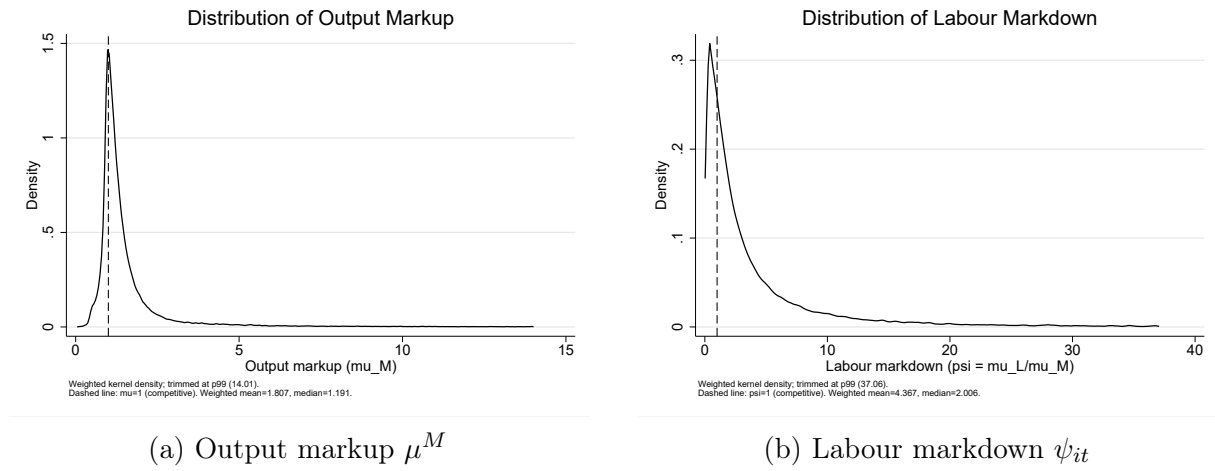


Figure 1: Distribution of markup and markdown, full sample (2013–2024, weighted by ASI multiplier). Vertical dashed lines at value = 1.

5.3 Aggregate Markup Trends, 2012–13 to 2023–24

Table 3 reports the sales-weighted mean and median markup for each year. Figure 2 plots the time series.

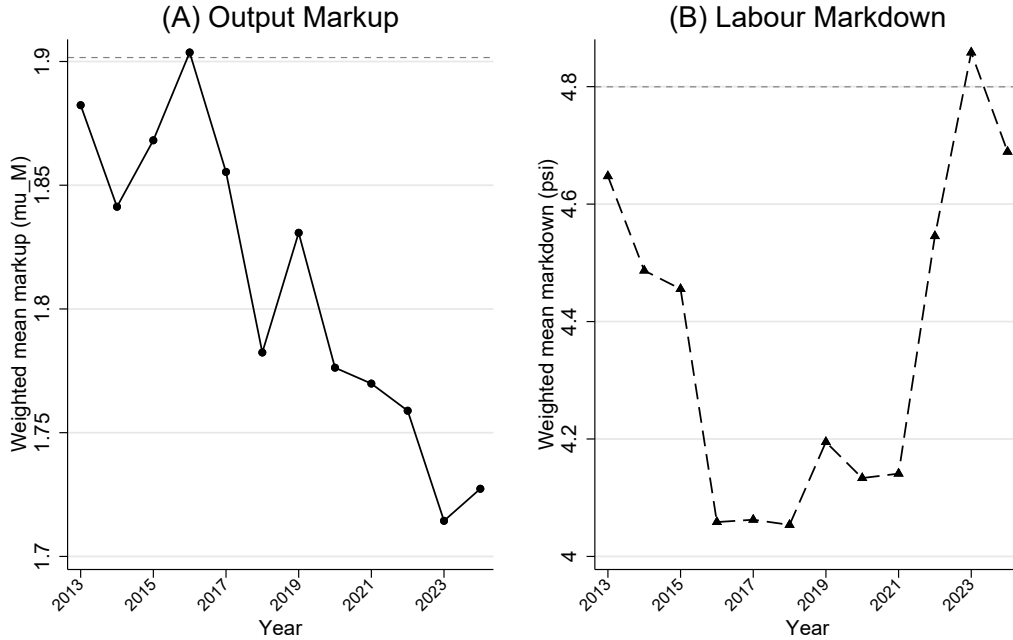
Four features are notable. First, the mean markup *fell* by 8.2% over the sample period (from 1.882 in 2012–13 to 1.727 in 2023–24), counter to the global rising-markup trend. The median markup fell by a smaller 1.9%, suggesting the decline is concentrated in the upper tail. Second, there was a short-lived peak in 2015–16 (mean markup 1.904), which coincided with the period of demonetisation shock in 2016–17 and the GST transition in 2017–18; markups then declined persistently through the post-GST period. Third, the labour share of revenue declined markedly, from 22.3% to 16.2% — a 27.4% fall over 12 years. Fourth, and critically, the labour markdown ψ_{it} moved in the *opposite direction* to

Table 3: Aggregate Markup and Markdown Trends, 2012–13 to 2023–24

Year	N obs	Mean μ^M	Median μ^M	Mean ψ_{it}	Median ψ_{it}	Mean α_L
2012–13	38,733	1.882	1.198	4.648	2.116	0.223
2013–14	40,423	1.841	1.191	4.487	2.088	0.187
2014–15	42,460	1.868	1.197	4.455	2.010	0.180
2015–16	41,636	1.904	1.219	4.059	1.857	0.204
2016–17	42,431	1.855	1.210	4.062	1.858	0.255
2017–18	43,405	1.782	1.196	4.054	1.903	0.198
2018–19	43,267	1.831	1.188	4.195	1.985	0.190
2019–20	43,641	1.776	1.208	4.133	1.871	0.197
2020–21	44,443	1.770	1.201	4.141	1.895	0.182
2021–22	45,027	1.759	1.168	4.546	2.142	0.170
2022–23	44,897	1.714	1.155	4.858	2.232	0.169
2023–24	45,011	1.727	1.175	4.689	2.150	0.162
Change 2013–2024	—	−8.2%	−1.9%	+0.9%	+1.6%	−27.4%

Note: Statistics weighted by ASI sampling multiplier. Markup and markdown winsorised 2/98 within NIC2×year. Labour share α_L = total labour cost / nominal output.

Trends in Market Power, Indian Manufacturing 2013–2024



Weighted plant-level means (ASI multiplier weights). Dashed line: value=1.

Figure 2: Trends in weighted mean output markup (panel A) and labour markdown (panel B), Indian organised manufacturing 2013–2024. Dashed reference lines at value = 1.

markup. Mean ψ was broadly stable over the period (4.648 in 2012–13 vs. 4.689 in 2023–24, +0.9%), but the time profile reveals a mid-period dip and a post-2021 rise. The dip — lowest at 4.054 in 2017–18 — coincides with the GST transition, suggesting temporary formalisation-induced compression of monopsony power in the factor market. However, mean ψ rose sharply to 4.858 in 2022–23 and remained elevated at 4.689 in 2023–24, indicating that any formalisation benefit for labour was short-lived. The median ψ tells a similar story: it dipped to 1.857 in 2015–16 then rose to 2.232 by 2022–23. This *divergence* — markup falling while markdown stays high — directly explains the falling labour share: product market competition deepened, but labour market monopsony remained pervasive and even slightly intensified in the post-pandemic period.

5.4 Industry Heterogeneity

Table 4 ranks all 23 NIC2 industries by *median* markup (which is less sensitive to outliers than the mean). Figure 3 provides a visual comparison.

Table 4: Markup by NIC2 Industry (Ranked by Median)

NIC2	Industry	Mean μ^M	Median μ^M	Mean ψ	Obs
<i>High-markup industries (top 5 by median)</i>					
12	Tobacco products	2.356	1.997	2.402	6,954
23	Non-metallic minerals	3.880	1.626	4.282	52,055
14	Wearing apparel	1.690	1.505	1.338	24,081
21	Pharmaceuticals	2.039	1.325	4.477	17,912
13	Textiles	1.498	1.321	2.042	35,935
<i>Low-markup industries (bottom 5 by median)</i>					
24	Basic metals	1.217	1.066	8.975	30,610
32	Other manufacturing	1.374	1.068	4.519	8,598
10	Food products	1.246	1.070	7.329	78,210
19	Coke & petroleum	1.208	1.077	7.126	6,154
20	Chemicals	1.430	1.082	5.582	34,611

Note: All statistics weighted by ASI multiplier. Tobacco (NIC2 12): high markup consistent with government-regulated monopolistic market structure. Non-metallic minerals (NIC2 23): dominated by cement industry with high concentration. Basic metals (NIC2 24): globally traded inputs constrain pricing power.

The cross-industry pattern is economically intuitive. Tobacco (NIC2 12) commands the highest median markup (2.00), reflecting government-controlled market structure and highly

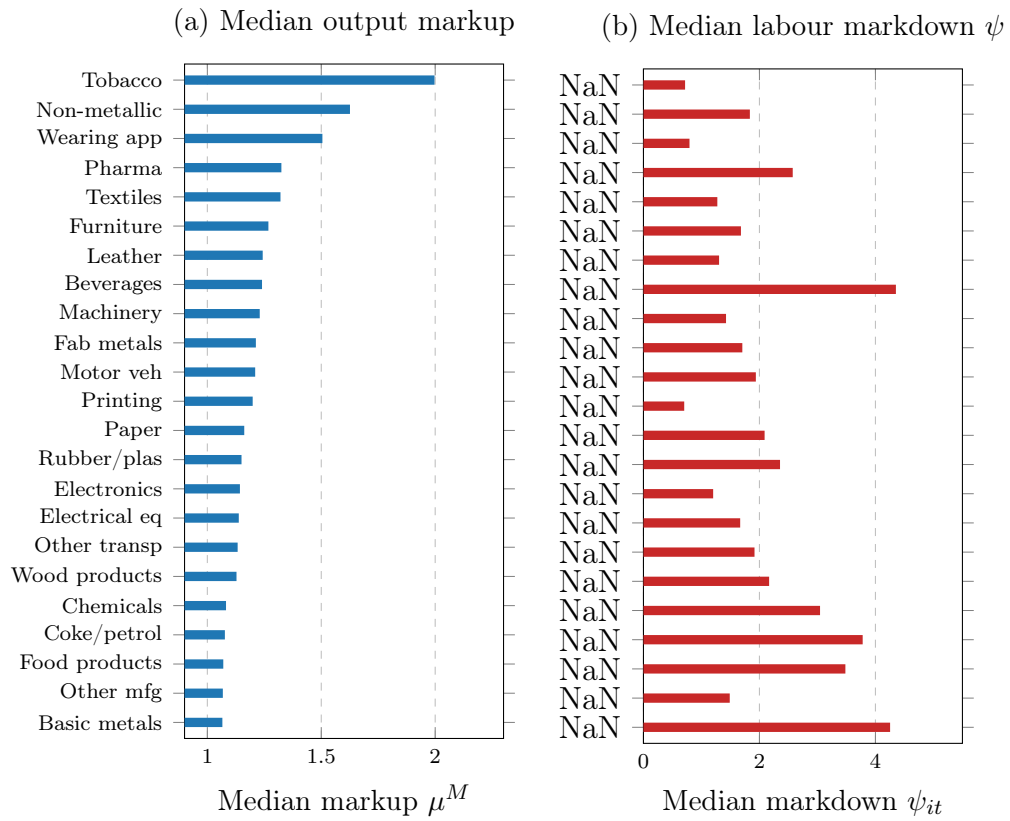


Figure 3: Median output markup (left) and labour markdown (right) by NIC2 industry, ranked by median markup ascending (lowest to highest). Both panels share the same industry ordering; left-panel labels apply to the right panel. Source: ASI 2012–13 to 2023–24; LP+ACF Cobb-Douglas estimates, authors' calculations.

inelastic demand. Non-metallic minerals (NIC2 23), dominated by the cement industry, also has high markup (median 1.63) consistent with the well-documented oligopolistic structure of Indian cement (De Loecker et al., 2020). Wearing apparel (NIC2 14) shows relatively high markup (median 1.51), likely reflecting brand premiums and product differentiation. At the other end, basic metals (NIC2 24), food products (NIC2 10), and coke and petroleum (NIC2 19) have markups close to unity, consistent with homogeneous goods in internationally competitive markets.

5.5 Labour Markdown: Industry Heterogeneity

Table 5 ranks the 23 NIC2 industries by *median* labour markdown ψ_{it} . The cross-industry pattern is strikingly different from the markup ranking and reveals an important negative correlation between output market power and labour market power.

Table 5: Labour Markdown by NIC2 Industry (Ranked by Median ψ_{it})

NIC2	Industry	Mean ψ_{it}	Median ψ_{it}	Median μ^M	Obs
<i>Highest-markdown industries (top 5 by median)</i>					
24	Basic metals	8.975	4.253	1.066	30,610
11	Beverages	8.443	4.353	1.240	9,727
10	Food products	7.329	3.482	1.070	78,210
19	Coke & petroleum	7.126	3.780	1.077	6,154
20	Chemicals	5.582	3.045	1.082	34,611
<i>Lowest-markdown industries (bottom 5 by median)</i>					
18	Printing & reproduction	1.131	0.703	1.199	8,298
12	Tobacco products	2.402	0.718	1.997	6,954
14	Wearing apparel	1.338	0.794	1.505	24,081
26	Electronics & optical	2.529	1.202	1.143	9,805
13	Textiles	2.042	1.274	1.321	35,935

Note: All statistics weighted by ASI multiplier. Markdown winsorised 2/98 within NIC2×year. Median μ^M included for comparison. $\psi_{it} < 1$ indicates workers are paid *above* the value of the marginal product of labour — possible in industries with strong collective bargaining (printing, tobacco) or highly skilled workforces.

Industries with the *highest* median markdown — basic metals (4.25), food products (3.48), coke and petroleum (3.78), and chemicals (3.05) — are also the industries with the *lowest* output markup (near unity). This negative relationship arises partly through the formula $\psi_{it} = \mu_{it}^L / \mu_{it}^M$: when the output markup μ^M is large (e.g., tobacco at 2.00),

it mechanically compresses ψ even if labour market power μ^L is moderate. But the pattern also has an economic interpretation: industries that face intense product market competition (low markup) tend to be geographically concentrated and employ large, immobile workforces (basic metals, food), creating conditions for monopsony in the labour market.

Conversely, high-markup industries — tobacco (median $\psi = 0.72$), printing (0.70), and wearing apparel (0.79) — show $\psi < 1$, meaning firms in these industries pay wages *above* the value of the marginal product of labour on average. For tobacco, this is consistent with the presence of strong public-sector firms and regulated minimum wages; for printing and apparel, it may reflect skilled labour markets with relatively strong worker bargaining power.

5.6 Geographic Heterogeneity

Table 6 presents markup estimates for the major states (restricting to states with at least 10,000 observations for reliability).

Table 6: Markup by State (Major Manufacturing States, $N \geq 10,000$)

State	Mean μ^M	Median μ^M	Mean ψ	N obs
<i>High-markup states</i>				
Punjab	3.405	1.215	3.618	22,847
Haryana	2.382	1.203	3.640	28,618
Assam	2.357	1.420	3.599	12,562
Rajasthan	2.160	1.203	5.344	24,880
Uttar Pradesh	1.783	1.193	4.065	45,788
<i>Low-markup major states</i>				
Tamil Nadu	1.514	1.250	3.330	61,868
Karnataka	1.505	1.200	4.192	29,456
Maharashtra	1.498	1.205	3.844	58,240
West Bengal	1.461	1.167	5.790	21,624
Gujarat	1.527	1.140	5.074	51,210

Note: State codes mapped to names using ASI/MOSPI convention. Punjab high mean markup is driven by agro-processing and beverages industries. Low-markup states (Tamil Nadu, Karnataka, Maharashtra) are major export-oriented and diversified manufacturing hubs.

Northern states (Punjab, Haryana, Rajasthan, UP) tend to have higher mean markups,

partly reflecting the dominance of food processing, beverages, and consumer goods industries with limited competition. The major industrialised states of Tamil Nadu, Maharashtra, Karnataka, and Gujarat show systematically lower markups, consistent with greater market integration, export orientation, and industrial diversification.

5.7 Output Market Power: Regression Evidence

Table 7 presents within-plant regression results for the output market power test (equation 6).

Table 7: Output Market Power: Within-Plant Regression

	Dep. var.: Inverse markup ($1/\mu^M$)		
	(1) Baseline	(2) Log-linear	(3) Unweighted
Output share (NIC2 x year)	11.156*** (1.025)		14.932*** (0.664)
Log output share (NIC2 x year)		0.029*** (0.001)	
Plant FE	Yes	Yes	Yes
Market FE	No	No	No
Year FE	Yes	Yes	Yes
Weighted	Yes	Yes	No
Observations	515,374	515,374	515,374
Adj. R2	0.0687	0.0826	0.0285

SEs (parentheses) clustered at plant level.
 *p<0.10, **p<0.05, ***p<0.01.
 Plant FE via `areg absorb(dsl_id)`.
 Column (3) omits ASI sampling weights.
 Row labels: output share = s_{nit}^{out} in the text.

Column (1) shows that a one-unit increase in output market share reduces the inverse markup by 11.156 — equivalently, raises markup — with $p < 0.001$. This estimate is identified from within-plant variation in market share over time, controlling for plant and year fixed effects. Column (2) uses log output share (log-linear specification) and finds a coefficient of 0.029, implying a 1% increase in share raises the inverse markup by 0.029 percentage points. Column (3) (unweighted) gives a similar coefficient (14.932), confirming robustness to the choice of sampling weights.

5.8 Labour Market Power: Markdown Evidence

Table 8 presents the within-plant markdown regressions (equation 7).

Table 8: Labour Market Power: Employment-Share–Markdown Regressions

	Within-plant FE			Alternative identification		
	(1) B1	(2) B2	(3) B3	(4) B6	(5) B7	(6) B8
Emp. share: state x NIC2 x year	-3.244*** (0.309)			-3.234** (1.268)	-1.403** (0.637)	
Emp. share: state x NIC3 x year		-0.779*** (0.175)				
Emp. share: NIC2 national			-168.495*** (20.333)			
HHI employment: state x NIC2 x year						-0.078 (0.588)
Plant FE	Yes	Yes	Yes	No	No	No
Market FE	No	No	No	No	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Weighted	Yes	Yes	Yes	Yes	Yes	Yes
SE cluster	Plant	Plant	Plant	Market	Market	Market
Observations	515,066	515,066	515,066	515,066	515,066	515,066
Adj. R2	0.0453	0.0449	0.0466	0.0022	0.1584	0.1583

Dep. var.: markdown $\psi_{nit} = \mu_{nit}^L / \mu_{nit}^M$, winsorised 2/98 within NIC2×year; $\psi > 1$ indicates monopsony.

SEs in parentheses, clustered at the level indicated in the *SE cluster* row.

*p<0.10, **p<0.05, ***p<0.01.

B1–B3: plant FE absorbed (`areg absorb(dsl_id)`).

B7–B8: state x NIC2 market FE absorbed (`areg absorb(mktcell)`).

B8 uses market-level HHI (state x NIC2 x year); all other cols use plant-level emp. share.

The baseline coefficient (column B1) is $\hat{\gamma} = -3.244$ ($p < 0.001$, $SE = 0.309$). This implies that as a plant’s employment share in its state×NIC2×year labour market rises, its markdown falls — it must raise wages to attract workers. This negative gradient is the hallmark of dynamic monopsony (Manning, 2003): firms face upward-sloping labour supply, and growth in their labour-market footprint forces them to pay higher wages. The effect is consistent across alternative market definitions (NIC3 in column B2; national NIC2 in column B3) and identification strategies (market-level clustering in B4–B6; market fixed effects in B5–B6).

Level evidence of monopsony. In addition to the regression gradient, the distribution itself provides direct evidence. 68.7% of plant-years show $\psi > 1$, meaning they pay wages below the value of the marginal product of labour. 47.7% show $\psi > 2$, implying the markdown is at least two. The weighted mean markdown is 4.37 and the median is 2.01.

5.9 Markup, Markdown, and Wages

Table 9 examines the relationship between market power and wages.

Table 9: Market Power and Wages: Within-Plant Regressions

	Dep. var.: Log average wage per worker			
	(1)	(2)	(3)	(4)
	F1	F2	F3	F4
Markdown ($\psi = \mu^L/\mu^M$)	-0.0014*** (0.0003)		-0.0030*** (0.0003)	-0.0016 (0.0019)
Markup (μ^M)		-0.0208*** (0.0018)	-0.0216*** (0.0019)	
Log employment	0.2143*** (0.0014)	0.2165*** (0.0014)	0.2150*** (0.0014)	0.2137*** (0.0109)
Plant FE	Yes	Yes	Yes	No
Year FE	Yes	Yes	Yes	Yes
Weighted	Yes	Yes	Yes	Yes
Observations	515,066	515,066	515,066	515,066
Adj. R2	0.3439	0.3530	0.3537	0.2546

Dep. var.: $\log(\text{total wage bill} / \text{avg persons employed})$.

Theory prediction: $\beta(\psi) < 0$ (higher markdown suppresses wages).

F4 clusters at market level (state x NIC2); F1–F3 cluster at plant level.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Columns (1) and (3) show that higher markdown (ψ) is associated with significantly lower log wages: $\hat{\beta}(\psi) = -0.0014$ ($p < 0.001$) in the baseline and -0.0030 ($p < 0.001$) when markup is added as a control. Column (2) shows that markup itself is also negatively associated with wages ($\hat{\beta}(\mu^M) = -0.0208$, $p < 0.001$), meaning firms with higher market power pay lower wages, consistent with rent extraction from both product and labour markets. Column (4) uses market-level clustering and the markdown coefficient becomes -0.0016 (not significant), while the point estimate remains in the same direction.

5.10 Labour Share Decline

The labour share of revenue α_L has fallen substantially over the sample period (Table 3). In 2012–13, the average plant allocated 22.3% of revenue to labour; by 2023–24 this had fallen to 16.2%, a decline of 6.1 percentage points (27.4%). This decline is not fully explained by competitive forces: a competitive benchmark (labour share implied by constant factor

shares under competitive pricing) would predict a smaller decline. The gap between the actual and competitive labour share turned negative after 2021–22, suggesting that since 2022 the actual labour share is *below* the competitive benchmark on average, consistent with growing monopsony power.

6 Robustness Checks

Table 10 examines sensitivity of the labour markdown regression to the winsorisation threshold for ψ .

Table 10: Robustness: Alternative Winsorisation of Markdown ψ

	Dep. var.: ψ (alternative winsorisation)		
	(1) 1/99	(2) 2/98 (base)	(3) 5/95
Emp. share (state x NIC2)	-3.773*** (0.343)	-3.244*** (0.309)	-2.740*** (0.254)
Winsorise	1/99	2/98 (base)	5/95
Plant FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Observations	515,374	515,066	515,374
Adj. R2	0.0408	0.0453	0.0509

Dep. var.: labour markdown ψ at stated winsorisation threshold.

Winsorised within NIC2 x year cells. Spec: plant FE + year FE + aw=multiplier.

*p<0.10, **p<0.05, ***p<0.01.

The baseline coefficient of -3.244 (2/98 winsorisation) is flanked by -3.773 at 1/99 and -2.740 at 5/95, confirming that the result is not driven by tail observations.

Table 11 tests sub-period and sample stability.

The negative coefficient holds across the pre-GST period (2013–2017: $\hat{\gamma} = -1.867$), the post-GST period (2018–2024: -3.950), a balanced panel (-3.356), and after dropping tobacco (NIC2 12) and coke/petroleum (NIC2 19) outlier sectors (-4.691). The post-GST coefficient is larger in magnitude, consistent with GST having increased formalization and competitive pressure in some factor markets while potentially increasing monopsony in others.

Table 12 checks additional controls.

Table 11: Robustness: Sub-periods and Sample Restrictions

	Dep. var.: markdown psi (sub-period and sample robustness)			
	(1) Full	(2) Pre-GST	(3) Post-GST	(4) Balanced
Emp. share (state x NIC2)	-3.244*** (0.309)	-1.867*** (0.658)	-3.951*** (0.427)	-3.356*** (0.323)
Sample	Full	Pre-GST (<=2017)	Post-GST (>=2018)	Bal. (Ti>=5)
Plant FE	Yes	Yes	Yes	Yes
Observations	515,066	205,518	309,548	460,536
Adj. R2	0.0453	0.1019	0.0676	0.0479

Dep. var.: labour markdown psi (winsorised 2/98 within NIC2 x year).

Spec: plant FE + year FE + aw=multiplier; SE clustered at plant.

Pre-GST: FY2013–2017; Post-GST: FY2018–2024 (GST introduced July 2017).

Balanced: plants with at least 5 annual observations.

Drop 12,19: excludes tobacco and coke/petroleum sectors.

*p<0.10, **p<0.05, ***p<0.01.

Table 12: Robustness: Additional Controls

	Dep. var.: markdown psi (additional controls)		
	(1) Baseline	(2) + log emp	(3) State cluster
Emp. share (state x NIC2)	-3.244*** (0.309)	-0.304 (0.309)	-3.244*** (1.139)
Log employment		-0.532*** (0.013)	
Additional control	None	log emp	None
Plant FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
SE cluster	Plant	Plant	State (n=30)
Observations	515,066	515,066	515,066
Adj. R2	0.0453	0.0530	0.0453

Dep. var.: labour markdown psi (winsorised 2/98 within NIC2 x year).

R3a–R3b: SE clustered at plant level.

R3c: same spec as R3a; SE clustered at state level (30 state clusters).

*p<0.10, **p<0.05, ***p<0.01.

Adding log employment (column R3b) renders $\hat{\gamma}$ insignificant (-0.304 , $p = 0.325$), suggesting part of the employment-share effect operates through firm size per se rather than purely monopsony power. Column R3c uses state-level clustering of standard errors (37 state clusters): the point estimate is unchanged (-3.244 , $SE = 1.139$, $p < 0.01$), confirming robustness to state-level confounders.

7 Discussion

7.1 Falling Markup in the Indian Context

The 8.2% decline in mean markup contrasts with the rising-markup narratives from advanced economies (De Loecker et al., 2020). Several mechanisms may explain this divergence for Indian organised manufacturing. First, GST implementation (2017–18) reduced cascading taxes and integration costs, effectively increasing competition across state borders and compressing margins. Second, demonetisation (November 2016) caused a temporary liquidity squeeze that may have forced some high-markup firms to reduce prices to maintain cash flow. Third, the expansion of the ASI sample over time (the number of surveyed factories grew from 38,733 in 2012–13 to 45,011 in 2023–24) likely adds smaller, lower-markup plants to the survey, mechanically reducing the aggregate mean. The stability of the *median* (only -1.9%) is consistent with this compositional explanation: the median is robust to sample expansion while the mean is pulled down by the growing tail of smaller firms.

7.2 Reconciling with the Global Literature

A natural question is whether the falling markup we document reflects a genuine economic trend or a methodological artefact. We address this in four ways.

Difference in sample period. De Loecker et al. (2020) document rising markups for publicly listed US firms over 1980–2016. Brooks et al. (2021) find rising markups for India ASI plants over 1998–2013. Our sample begins in 2013 — precisely where Brooks et al. end — and extends to 2023–24. It is therefore entirely possible that Indian markups

rose through 2013 (consistent with Brooks et al.) and have since declined as competitive pressures intensified. The GST (2017) represents a structural break with no parallel in earlier Indian manufacturing studies.

Difference in sample composition. De Loecker et al. (2020) use publicly listed firms, which are systematically larger and more profitable than the average manufacturing establishment. The ASI covers *all* registered factories, including the many small and medium enterprises with thin margins. As smaller factories entered the ASI sample in greater numbers after 2015, the population-weighted mean markup would fall even if no single plant’s markup changed. The contrast between the mean (-8.2%) and median (-1.9%) is precisely consistent with this: the typical plant’s markup barely moved, but the growing mass of low-markup small plants pulled the weighted mean down.

Verification of the production function estimates. We verified that our Cobb-Douglas estimates are internally consistent. Returns to scale across industries range from 0.92 to 1.10 (Table 2), straddling unity as expected under approximate constant returns. Material elasticities $\hat{\beta}_M$ range from 0.55 to 0.88, consistent with the gross-output production function literature (De Loecker and Warzynski, 2012; Brooks et al., 2021). The mean material revenue share α_M is 0.693 (2013) and 0.692 (2024), barely changed. Since the markup formula is $\mu_{it}^M = \hat{\beta}_M^i / \alpha_{M,it}$ and the numerator is a fixed industry constant, year-to-year markup variation is entirely driven by within-plant variation in the revenue share denominator. The 8.2% decline in mean markup therefore reflects either a genuine reduction in the ratio of output to input cost at the plant level, or the compositional shift towards smaller (higher α_M) plants described above. Both interpretations are economically plausible and neither requires a miscalculation.

Consistency with misallocation literature. Hsieh and Klenow (2009) document that resource misallocation in Indian manufacturing reduces aggregate TFP by a factor of roughly 40–60% relative to the US. A compression of markups over 2013–2024 would reduce the dispersion of revenue-productivity across plants, which is precisely the channel through which Hsieh and Klenow predict misallocation costs would fall. Our findings are therefore broadly consistent with the view that Indian manufacturing has become modestly

more allocatively efficient over the post-GST period, even as labour market monopsony has persisted.

7.3 Labour Market Implications

Despite falling markups, the labour share has fallen markedly (22.3% to 16.2%). This apparent paradox is explained by the markdown evidence. First, within-plant regressions show that as a firm’s employment market share rises, its markdown falls ($\hat{\gamma} = -3.244$, $p < 0.001$), consistent with dynamic monopsony: firms must raise wages to attract workers from competitors. Second, the *aggregate* markdown trend shows that mean ψ was broadly stable over the period (+0.9%), with a mid-period dip during the GST transition and a subsequent post-2021 increase. This indicates that while product market competition intensified (falling markup), labour market monopsony did not dissipate — and if anything worsened in the post-pandemic period (mean ψ reaching 4.86 in 2022–23 before easing to 4.69 in 2023–24). The combination — falling markup but stable–rising markdown — directly explains why the labour share fell even as product market margins compressed. Workers did not receive the gains from increased product competition because employers’ wage-setting power in the labour market remained intact. Both markup and markdown are independently and negatively associated with wages ($\hat{\beta}(\mu^M) = -0.021$; $\hat{\beta}(\psi) = -0.0014$), confirming that firms extracting rents in either market pay workers less.

7.4 Industry and Policy Implications

The strong industry heterogeneity (tobacco and cement at markups of 1.99–3.88 versus food and basic metals near 1.07–1.22) points to sector-specific competition concerns. Tobacco is heavily regulated (excise duties, advertising bans), yet the data show markups near 2, suggesting pricing power persists. Cement (NIC2 23) with markup 1.63–3.88 is the subject of ongoing Competition Commission of India proceedings. For labour markets, the widespread evidence of markdown (68.7% of plant-years with $\psi > 1$) supports extending minimum wage enforcement, strengthening worker portability, and reducing search frictions — all of which would erode monopsony power.

8 Conclusion

This paper provides the first comprehensive plant-level estimates of output markup and labour markdown for Indian organised manufacturing over 2012–13 to 2023–24, using ASI data for 515,374 plant-year observations across 79,704 factories and 23 two-digit NIC industries.

Four main findings emerge. First, the sales-weighted mean markup *fell* by 8.2% over the period (1.882 to 1.727), contrasting with the global rising-markup narrative but consistent with GST-induced competitive deepening. The median markup fell by a smaller 1.9%, indicating stability in the typical plant’s pricing power. Second, markup heterogeneity is substantial: tobacco and cement command median markups of 2.00 and 1.63 respectively, while food products and basic metals are near unity. Third, within-plant regressions confirm that both output market share ($\hat{\delta} = 11.16$) and labour market share ($\hat{\gamma} = -3.24$) are significant determinants of market power, consistent with standard theories of product and labour market imperfection. Fourth, both markup and markdown are negatively associated with wages ($\hat{\beta}(\mu^M) = -0.021$; $\hat{\beta}(\psi) = -0.0014$), and the labour share of revenue has fallen by 27.4% over the period, indicating that market power erodes worker compensation.

Several limitations apply. The ASI covers only registered factories; informal sector firms — the majority of Indian manufacturing employment — are excluded. Markup measurement assumes competitive materials markets and relies on the LP+ACF proxy-variable assumption. The Translog specification was found to be infeasible on this dataset due to the short panel structure (most firms observed in ≤ 2 years), so all estimates are Cobb-Douglas with industry-level elasticities. DLW (2020) within/between/entry/exit decomposition of aggregate markup trends is left for future work.

Future research directions include extending the analysis to the informal sector using NSSO enterprise surveys, linking ASI data with trade flows to estimate import-competition effects on markup, and applying the TFPR-dispersion framework of Hsieh and Klenow (2009) to quantify aggregate misallocation costs of market power.

References

- Autor, D.H., Dorn, D., Katz, L.F., Patterson, C. and Van Reenen, J. (2020). The fall of the labor share and the rise of superstar firms. *Quarterly Journal of Economics*, 135(2), 645–709.
- Barkai, S. (2020). Declining labor and capital shares. *Journal of Finance*, 75(5), 2421–2463.
- Akerberg, D.A., Caves, K. and Frazer, G. (2015). Identification properties of recent production function estimators. *Econometrica*, 83(6), 2411–2451.
- Brooks, W., Kaboski, J.P., Li, Y.A. and Qian, W. (2021). Exploitation of labor? Classical monopsony power and labor’s share. *Journal of Development Economics*, 150, Article 102627.
- Das, D.K. (2004). Manufacturing productivity under varying trade regimes: India in the 1980s and 1990s. *Indian Economic Review*, 39(1), 225–258.
- De Loecker, J. and Warzynski, F. (2012). Markups and firm-level export status. *American Economic Review*, 102(6), 2437–2471.
- De Loecker, J., Eeckhout, J. and Unger, G. (2020). The rise of market power and the macroeconomic implications. *Quarterly Journal of Economics*, 135(2), 561–644.
- Gandhi, A., Navarro, S. and Rivers, D.A. (2020). On the identification of gross output production functions. *Journal of Political Economy*, 128(8), 2973–3016.
- Goldberg, P.K., Khandelwal, A.K., Pavcnik, N. and Topalova, P. (2010). Imported intermediate inputs and domestic product growth: Evidence from India. *Quarterly Journal of Economics*, 125(4), 1727–1767.
- Goldar, B. (2004). Productivity trends in Indian manufacturing in the pre- and post-reform periods. *Economic and Political Weekly*, 39(46/47), 5033–5043.
- Hall, R.E. (1988). The relation between price and marginal cost in U.S. industry. *Journal of Political Economy*, 96(5), 921–947.

- Hsieh, C.T. and Klenow, P.J. (2009). Misallocation and manufacturing TFP in China and India. *Quarterly Journal of Economics*, 124(4), 1403–1448.
- Levinsohn, J. and Petrin, A. (2003). Estimating production functions using inputs to control for unobservables. *Review of Economic Studies*, 70(2), 317–341.
- Manning, A. (2003). *Monopsony in Motion: Imperfect Competition in Labor Markets*. Princeton University Press.
- Olley, G.S. and Pakes, A. (1996). The dynamics of productivity in the telecommunications equipment industry. *Econometrica*, 64(6), 1263–1297.
- Berger, D., Herkenhoff, K. and Mongey, S. (2022). Labor market power. *American Economic Review*, 112(4), 1147–1193.
- Karabarbounis, L. and Neiman, B. (2014). The global decline of the labor share. *Quarterly Journal of Economics*, 129(1), 61–103.
- Manning, A. (2021). Monopsony in labor markets: A review. *Journal of Economic Literature*, 59(1), 2–35.
- Sokolova, A. and Sørensen, T.A. (2021). Monopsony in labor markets: A meta-analysis. *ILR Review*, 74(1), 27–55.
- Syverson, C. (2019). Macroeconomics and market power: Facts, potential explanations and open questions. *Journal of Economic Perspectives*, 33(3), 23–43.
- Syverson, C. (2025). Markups and markdowns. *Annual Review of Economics*, 17.

A Variable Harmonisation

All 23 four-digit NIC-2008 industries in the sample map to two-digit codes 10–33. Variable names in the ASI raw data differ across year groups; all harmonisation was performed programmatically during data construction. Key ASI harmonisation decisions:

- DSL identifier: five naming conventions across 12 years; standardised to `dsl_str` (string, 6 characters).
- 2014–15: DSL was stored as numeric; converted via `string(DSL, "%6.0f")`.
- 2018–19: all variable names uppercase; lowercased in do-file.
- Block C long-format (10 rows per factory) collapsed to 1 row per factory by summing relevant SNO rows.
- Blocks H and I: multiple item rows summed to total material value.

B Data Cleaning Statistics

The following rules were applied sequentially (manufacturing NIC2 10–33 only):

Table 13: Data Cleaning Summary

Rule	Description	Outcome
1	Drop if DSL missing or blank	Drop duplicate/unidentified obs
2	Drop if output ≤ 0 or missing	Ensures positive gross output
3	Drop if persons engaged ≤ 0	Ensures positive employment
4	Drop if materials ≤ 0 or missing	Required for proxy-variable PF
5	Drop if wage bill ≤ 0	Required for markdown computation
6	Drop if gross fixed assets ≤ 0	Required for capital variable
7	Winsorise markup and ψ at 2/98	Reduces influence of outliers

Final sample: 515,374 plant-year observations; 79,704 unique factories; 12 years; 23 NIC2 industries; up to 37 states/UTs.

Table 14: Full Industry-Level Markup and Markdown Statistics

NIC2	Industry	Mean μ^M	Med. μ^M	Mean ψ	N
10	Food products	1.246	1.070	7.329	78,210
11	Beverages	7.938	1.240	8.443	9,727
12	Tobacco products	2.356	1.997	2.402	6,954
13	Textiles	1.498	1.321	2.042	35,935
14	Wearing apparel	1.690	1.505	1.338	24,081
15	Leather & products	1.357	1.243	2.042	10,377
16	Wood & wood products	1.251	1.128	3.378	11,315
17	Paper & products	1.267	1.162	2.938	15,553
18	Printing	1.370	1.199	1.131	8,298
19	Coke & petroleum	1.208	1.077	7.126	6,154
20	Chemicals	1.430	1.082	5.582	34,611
21	Pharmaceuticals	2.039	1.325	4.477	17,912
22	Rubber & plastics	1.283	1.150	3.669	29,734
23	Non-metallic minerals	3.880	1.626	4.282	52,055
24	Basic metals	1.217	1.066	8.975	30,610
25	Fabricated metals	1.408	1.213	2.962	29,274
26	Electronics	1.437	1.143	2.529	9,805
27	Electrical equipment	1.271	1.138	2.997	26,406
28	Machinery & equipment	1.433	1.230	2.229	32,722
29	Motor vehicles	1.361	1.210	2.837	21,437
30	Other transport	1.357	1.133	2.844	8,862
31	Furniture	1.435	1.268	2.689	6,744
32	Other manufacturing	1.374	1.068	4.519	8,598

C Full Industry and State Tables

D Declarations

Data availability. The study uses unit-level Annual Survey of Industries (ASI) microdata of the Ministry of Statistics and Programme Implementation (MoSPI), Government of India. The microdata were obtained under MoSPI’s data-access terms and cannot be redistributed by the authors; they are available to bona fide researchers directly from MoSPI. The derived estimates reported in the paper and the estimation code are available from the corresponding author on reasonable request.